Step 2: Multicollinearity and Redundancy Diagnostics

This step diagnoses the full predictor universe used in Step 1. It does not decide the final variable set by itself. It flags where the same concept is probably being counted more than once.

Headline Findings

- Predictor count checked: 70
- High-correlation pairs with |r| >= 0.70: 51
- Near duplicate or inverse pairs with |r| >= 0.95: 10
- Variables with VIF >= 10: 43
- Variables with VIF >= 5: 51

Highest VIF Variables

Variable	Family	VIF
block2_apartment_lt5_storeys_count	housing_form_density	inf
block2_apartment_total_count	housing_form_density	inf
block5_no_car_household_share	transportation_access	inf
block5_transit_commute_share_preferred	transportation_access	inf
block5_tts_no_car_household_share	transportation_access	inf
block5_tts_transit_trip_share	transportation_access	inf
block4_mayoral_top_two_margin	mayoral_competitiveness	inf
block2_apartment_duplex_count	housing_form_density	inf
block2_apartment_5plus_storeys_count	housing_form_density	inf
block4_mayoral_winner_margin	mayoral_competitiveness	inf
block2_renter_share	housing_tenure	8100.055
block2_owner_share	housing_tenure	8033.196
block2_structural_type_total_dwellings	housing_form_density	284.495
block2_condo_status_total_dwellings	housing_form_density	259.010
block4_federal_vote_fragmentation	federal_competitiveness	163.980
block4_effective_federal_parties_5pct	federal_competitiveness	154.937
block1_age_65_plus_share	age_structure	146.305
block1_age_18_34_share	age_structure	129.804
block4_provincial_vote_fragmentation	provincial_competitiveness	99.928
block4_effective_provincial_parties_5pct	provincial_competitiveness	89.222

Strongest Correlated Pairs

A	B	r	Flag
block5_no_car_household_share	block5_tts_no_car_household_share	1.000	near_duplicate_or_inverse
block5_transit_commute_share_preferred	block5_tts_transit_trip_share	1.000	near_duplicate_or_inverse
block4_mayoral_top_two_margin	block4_mayoral_winner_margin	1.000	near_duplicate_or_inverse
block2_renter_share	block2_owner_share	-1.000	near_duplicate_or_inverse

A	B	r	Flag
block2_structural_type_total_dwellings	block2_condo_status_total_dwellings	0.994	near_duplicate_or_inverse
block4_effective_federal_parties_5pct	block4_federal_vote_fragmentation	0.994	near_duplicate_or_inverse
block4_effective_provincial_parties_5pct	block4_provincial_vote_fragmentation	0.989	near_duplicate_or_inverse
block2_population_density_per_km2	block2_apartments_per_km2	0.971	near_duplicate_or_inverse
block4_effective_mayoral_candidates_5pct	block4_mayoral_vote_fragmentation	0.967	near_duplicate_or_inverse
block2_apartment_5plus_storeys_count	block2_apartment_total_count	0.960	near_duplicate_or_inverse
block2_same_address_1yr_share	block2_same_address_5yr_share	0.928	strong
block2_structural_type_total_dwellings	block2_apartment_total_count	0.924	strong
block2_apartment_total_count	block2_condo_status_total_dwellings	0.923	strong
block1_age_65_plus_share	block1_median_age	0.918	strong
block3_immigrant_share	block3_non_official_mother_tongue_share	0.916	strong
block3_non_citizen_share	block3_citizen_adult_share	-0.913	strong
block3_recent_immigrant_share	block3_non_citizen_share	0.899	strong
block3_immigrant_share	block3_visible_minority_share	0.889	strong
block2_apartment_5plus_storeys_count	block2_condominium_dwellings_count	0.876	strong
block2_apartment_5plus_storeys_count	block2_condo_status_total_dwellings	0.865	strong
block2_structural_type_total_dwellings	block2_apartment_5plus_storeys_count	0.865	strong
block2_apartment_share	block2_detached_share	-0.856	strong
block2_apartments_per_km2	block2_condos_per_km2	0.851	strong
block2_apartment_total_count	block2_condominium_dwellings_count	0.844	moderate_high
block1_age_18_34_share	block2_same_address_1yr_share	-0.827	moderate_high

Plain-Language Redundancy Summary

The most important high-correlation or inverse relationships are no car household share and TTS no car household share with correlation 1.000; transit commute share and TTS transit trip share with correlation 1.000; mayoral top two margin and mayoral winner margin with correlation 1.000; renter share and owner share with correlation -1.000; structural type total dwellings and condo status total dwellings with correlation 0.994; effective federal parties above five percent and federal vote fragmentation with correlation 0.994; effective provincial parties above five percent and provincial vote fragmentation with correlation 0.989; population density and apartments per square kilometre with correlation 0.971; effective mayoral candidates above five percent and mayoral vote fragmentation with correlation 0.967; apartment five plus storeys count and apartment total count with correlation 0.960. These are the relationships most likely to double-count the same social concept if they are fed into dimension reduction together.

Family-Level Redundancy

Family	Vars	Max	r	
mayoral_competitiveness	5	1.000		
transportation_access	5	1.000		
housing_tenure	2	1.000		
housing_form_density	14	0.994		
federal_competitiveness	4	0.994		
provincial_competitiveness	4	0.989		
residential_stability	2	0.928		
age_structure	4	0.918		

Family	Vars	Max	r	
immigration_citizenship	7	0.916		
service_access	11	0.808		
service_contact	8	0.679		
household_class	4	0.677		

Interpretation

The diagnostics confirm Zack's warning: the modelling table contains variables that are not merely correlated but sometimes structurally tied. Tenure is the clearest example because renter and owner shares are inverse codings of the same concept. Housing form and density also form a dense cluster, especially because apartment counts, apartment shares, condo variables, density variables, and total dwelling counts partly track the same urban form.

VIF is stricter than pairwise correlation because it asks whether a variable can be reconstructed from all other predictors together. High VIF does not mean the variable is unimportant; it means the model cannot cleanly separate its unique contribution from nearby variables. That is why Step 3 should use family comparisons rather than deleting variables mechanically by VIF.

The practical conclusion is that the final cleaned model should choose representatives from correlated families, then test substitutions and connected-family combinations. That preserves the idea that A may work better with D than with C, which pairwise correlation alone cannot discover.